



Vidyavardhini's College of Engineering and Technology

Department of Artificial Intelligence & Data Science

AY: 2025-26

Class:	BE	Semester:	VIII
Course Code:		Course Name:	RS

Name of Student:	BARI ANKIT VINOD
Roll No. :	61
Assignment No.:	4
Title of Assignment:	
Date of Submission:	
Date of Correction:	

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Completeness	5	5
Demonstrated Knowledge	3	2
Legibility	2	2
Total	10	9

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Completeness	5	3-4	1-2
Demonstrated Knowledge	3	2	1
Legibility	2	1	0

Checked by

Name of Faculty : Mr. Ramesh Joshi

Signature :

Date :

1) You are developing a travel recommendation system. A user says "I want a peaceful place with cultural heritage." How can knowledge graphs and reasoning rules be used to recommend destinations that match this preference, even if the user doesn't mention specific locations?

→ To handle a vague query like "peaceful place with cultural heritage", a knowledge-based system can infer suitable destinations using structured relationships and rules.

S1 - Build a knowledge graph -

- Nodes - destination (e.g., cities, monuments, regions)
 - Attributes (peaceful, heritage, spiritual, crowded)
- edges - Jaipur → has attribute → cultural heritage.
 - Rishikesh → has attribute → peaceful
 - Hampi → has attribute → cultural heritage + peaceful

S2 - define reasoning rules -

- If a place has heritage sites and low tourist density → classify as peaceful cultural destination.
- If a place is a UNESCO site → infer cultural heritage
- If a place is a spiritual town or nature retreat → infer peaceful.

S3 - Query expansion + inference.

- User query: peaceful + cultural heritage.
- System - expands "peaceful" → low crowd, nature, spiritual
 - expands "cultural heritage" → historical monuments.

S4 - Recommendations -

- Hampi (heritage + calm environment)
- Khajuraho (historical + less crowded)
- Varanasi (spiritual + heritage)
- Pondicherry (cultural, vibe)

Q. 2) A Knowledge-based recommendation system utilizes a knowledge base containing 100 domain specific rules to generate recommendations. In a test scenario, it achieved an average precision of 0.90 and recall 0.80, meanwhile, a collaborative filtering system achieved an average precision of 0.85 and recall of 0.85. If the dataset comprises 2000 users and 1000 items, calculate the coverage of each system, defined as the proportion of items in the dataset that received at least one recommendation, which system offers better coverage, and by how much?

→ given, users = 2000 precision / recall - P = 0.90 R = 0.80
 items = 1000 P = 0.85 R = 0.85

S1. Interpret coverage -

$$\text{coverage} = \frac{\text{No. of unique recommendation items}}{\text{Total items}}$$

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- we approximate -
- Higher recall $\rightarrow$ more items recommended overall
- Higher diversity $\rightarrow$ higher coverage.

S2. Estimate number of recommended items.

- Assume - Total relevant items proportional across systems.
- Use recall to estimate how many items are retrieved.
- knowledge-based system -

$$\text{coverage}_{kb} \approx 0.80$$

- collaborative filtering -

$$\text{coverage}_{cf} \approx 0.85$$

S3. Compare coverage -

- collaborative filtering - 85%

- knowledge-based - 80%

FOR EDUCATIONAL USE

$$\therefore 0.85 - 0.80 = 0.05 \quad (5\%)$$